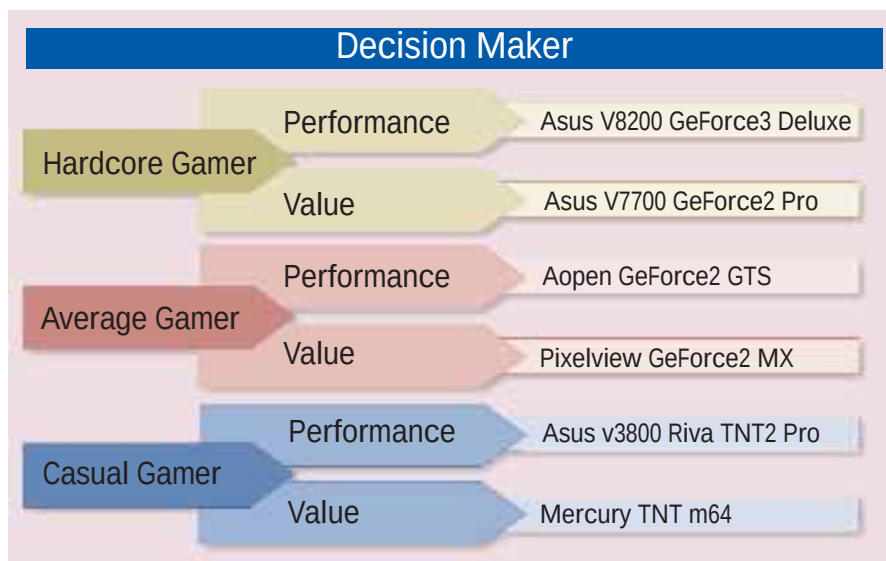




What all this means is that if you have a high-end 3D card but lack a good processor (you need at least an 800 MHz P-III with 128 MB of RAM), then you will experience a huge performance loss, and you won't be able to experience more than half of what the card is capable of—you can play a high-end game with 32 bit textures and at a fairly high resolution of 1024x768 even on a low-end machine, but minus the effects.

As we move on to the Matrox range of cards, we found that the performance of these cards varied in the Direct 3D tests but the performance in the OpenGL tests was more or less the same. The Matrox G450 32MB logged a score of 1276 points in 3DMark2000 which was a lot lower than what we recorded on the high-end machine. The same was the case in the 3DMark2001 test where we recorded a score of only 470 points: a lot lower than the score of 1183 recorded on our high-end machine. In the *Quake III* test the card logged an average of 46.7 fps in the normal mode, which was much lower than the score of 73 fps recorded on the high-end machine. But there wasn't any noticeable difference in the scores logged with high-quality and max-quality settings. What this indicates is that with all the effects enabled, this card would not give you a performance boost even if you have a high-end machine but if you don't mind playing games with the extra visual goodies turned off and at a lower resolution, then you will get a better performance.

The Elsa Synergy3 Quadro2 MXR is based on the nVidia Quadro 2MXR chipset. This card scored 2790 points in 3DMark 2000, which is almost half the score recorded on the high-end machine. This card is mainly useful for working with

high-end 3D packages such as 3D Max and Maya where you also have to possess a high-end machine. Therefore this card is not recommended for anybody possessing a P-III 1.2GHz or lower. If you want the best performing M64-based card then you should go in for the Mercury TNT M64. This card was the best in the shootout with acceptable performance. In the 3DMark 2000 test it returned a score of 1298, which again was lower than its score on the high-end machine. In the *Quake III* test it gave a playable frame rate of 46 fps in the low-quality mode, but with the max settings on, the card died out with an average 7.1 fps. The comparative scores of this card on both the machines show that it will not give you a huge performance boost if you use it with a high-end machine. This card is recommended for the casual gamer who does not possess a high-end machine.

What about the extra goodies?

These days most of the 3D cards come with a lot of extra goodies, besides the inherent features of the card itself. Take for example the Asus V8200 GeForce3 Deluxe which sits right on the top of the list with a total package that comes loaded with VR glasses, game CDs, utilities and the manual and driver CD. The packaging of this product is excellent. In fact, all the products that we received from Asus were good as far as packaging is concerned and some were exceptionally good such as the GeForce3. The Matrox Marvel G450 ETV also came with good packaging and included the cables for connecting dual monitors to the card. Elsa Synergy3 Quadro2 MXR contained the DVI/VGA adapter along with detailed manuals.

CONCLUSION

This graphics card test is more or less a clear representation of what is available in the market today. It wouldn't be unfair to say that nVidia rules the market. From the extremely high-end solution to the low-end 'will get your game running' cards—all have nVidia written all over them. The consumer is more or less swamped with solutions in the mid-range price bracket (Rs 2000-8000). Here, one should bear in mind that apart from the extra goodies (free games, software and other accessories) there isn't much to separate the cards.

As far as pure performance is concerned there is absolutely no



Asus V8200 GeForce3 Deluxe



Aopen PA256 GeForce2 MX

doubt as to who the winner is. The **Asus V8200 GeForce3 Deluxe** beats everyone hollow. It is simply hard not to be awestruck with its amazing processing power. Sadly, this card costs more than most PCs and needs a souped up CPU to get the best out of it. Hence, it can be recommended only to the extreme gamer/enthusiast.

For lesser mortals there is a wide range to choose from. The **AOpen PA256 GeForce2 MX** is the clear winner as far as the value segment is concerned. It also packs in a decent punch and will run most games at low to medium resolutions. 